Course Syllabus – Spring 2026



Course	CS 3345 - 501
Course Title	Data Structures & Algorithm Analysis
Term	Spring 2026
Meetings	Tu/Th 5:30 - 6:45 PM (ECSS 2.306)

Professor Contact Information

Dr. Omar Hamdy
Email: omar.hamdy@utdallas.edu
Tel: 972-883-2692
Office: ECSS 3.206

Office Hours:

Fri 3:00 – 5:00 PM
Or by appointment

Course Pre-requisites, Co-requisites, and/or Other Restrictions

Pre-requisites:

CE 2305 or CS 2305 or TE 2305 with a grade of C or better **and**
CE 2336 or CS 2336 or TE 2336 with a grade of C or better

Co-requisites:

CS 3341 or SE 3341 or ENGR 3341

Course Description

CS 3345 - Data Structures and Introduction to Algorithmic Analysis (3 semester credit hours) Analysis of algorithms including time complexity and Big-O notation. Analysis of stacks, queues, and trees, including B-trees. Heaps, hashing, and advanced sorting techniques. Disjoint sets and graphs. Course emphasizes design and implementation.

Student Learning Objectives/Outcomes

CLO 1	Ability to use/analyze Asymptotic notations, recurrences, algorithm analysis
CLO 2	Ability to use/analyze Lists, stacks, queues, hashing, priority queues
CLO 3	Ability to use/analyze Binary search trees, Balanced binary search trees

CLO 4	Ability to use/analyze Graphs, Depth-first search, Topological ordering
CLO 5	Ability to use/analyze Breadth-first search, Dijkstra's algorithm
CLO 6	Ability to use/analyze Algorithms of Prim and Kruskal, Disjoint-set Union-Find problem

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Required Textbooks and Materials

- Data Structures and Algorithm Analysis in Java - Third Edition (2011), Mark Weiss, Addison-Wesley, ISBN: 978-0-13-257627-7
- [OPTIONAL]: OpenDas, CS3 Data Structures & Algorithms
- JAVA compiler IDE of your choice: Eclipse (Recommended), BlueJ, JDeveloper, NetBeans, etc

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Grading Policy

Coursework Weight

Assignment	Weight
Homework 1	10%
Homework 2	10%
Homework 3	10%
Homework 4	10%
Exam 1	20%
Exam 2	20%
Final Exam	20%

Above could change during the semester

Letter Grade Chart

Score	Grade
Above 97	A+
93-96	A
90-92	A-
87-89	B+
83-86	B
80-82	B-
70-79	C
60-69	D
Below 60	F

Assignments & Academic Calendar

Exams

Exams will be a combination of simple problems, definitions, and diagram problems. Exam materials will be taken mainly from classroom lectures and textbook covered chapters and require a clear understanding of topics discussed in class. All exams will be individual work.
Final Exam is a comprehensive exam that is common across all CS3345 sections.

Assignments

Homework will consist of problems that will need to be worked out individually by each student. Each student will solve their homework on their own. Collaboration is not allowed.

Course Schedule

This is a **tentative** schedule which will be adjusted during the semester to accurately meet the class CLOs.

Week	Lecture Date	Material Covered
1	Jan. 20, 22	Class Intro, Chapter 1
2	Jan. 27, 29	Chapter 2, Chapter 3
3	Feb. 03, 05	Chapter 3
4	Feb. 10, 12	Chapter 3, Chapter 4
5	Feb. 17, 19	Chapter 4
6	Feb. 24, 26	Chapter 4, Chapter 5
7	Mar. 03, 05	Chapter 5
8	Mar. 10, 12	Chapter 6, Exam 1
Mar. 16 – Mar 22, Spring Break		
10	Mar. 24, 26	Chapter 6
11	Mar 31, Apr 02	Chapter 7
12	Apr. 07, 09	Chapter 7, Chapter 8
13	Apr. 14, 16	Chapter 8, Exam 2
14	Apr. 21, 23	Chapter 9
15	Apr. 28, 30	Chapter 9
16	May. 05, 07	Chapter 9, Final Exam

Course & Instructor Policies

Late Work

Assignments **will not** be accepted late. Students are expected to have read the instructions and to know the time an assignment is due. If eLearning/Blackboard has an error, and the instructor is informed, an extension will be granted; otherwise, homework is due when it is due.

Makeup Exams

- Students who miss an exam or test must have a valid reason for this. The situation should be clearly described in an e-mail to Dr. Hamdy directly. For medical issues, a doctor's note is the required form of proof. If you are sick, please go to the doctor and get a note. A dental appointment or other non-emergency health situation is not an acceptable excuse for missing an examination.
- If a student sits any exam, this means the student accepts the responsibility for that exam. Once taken, the exam will not be given again, and no make-up will be scheduled.

Communication

I do not read eLearning/Blackboard e-mails. Please use my UTD e-mail account I liked in contact information section above for any communications.

Class Attendance

Attendance is mandatory for this class in accordance with CS Department policies. Furthermore, your attendance record may be considered as a factor in calculating any potential curve or bonus applied at the end of the semester.

Ethical Behavior

In addition to the university policies and disciplinary actions described under the Academic Integrity section below, Dr. Hamdy's response to any form of dishonesty is a score of **0** on the entire assignment. Any students involved will, according to Departmental Policy, be referred to the Director of Community Standards & Conduct for adjudication.

Comet Creed

This creed was voted on by the UT Dallas student body in 2014. It is a standard that Comets choose to live by and encourage others to do the same:

"As a Comet, I pledge honesty, integrity, and service in all that I do."

Student Conduct & Discipline

The University of Texas System and The University of Texas at Dallas have rules and regulations for the orderly and efficient conduct of their business. It is the responsibility of each student and each student organization to be knowledgeable about the rules and regulations which govern student conduct and activities. General information on student conduct and discipline is contained in the UTD publication, *A to Z Guide*, which is provided to all registered students each academic year.

The University of Texas at Dallas administers student discipline within the procedures of recognized and established due process. Procedures are defined and described in the *Rules and Regulations, Board of Regents, The University of Texas System, Part 1, Chapter VI, Section 3*, and in Title V, Rules on Student Services and Activities of the university's *Handbook of Operating Procedures*. Copies of these rules and regulations are available to students in the Office of the Dean of Students, where staff members are available to assist students in interpreting the rules and regulations (SU 1.602, 972/883-6391).

A student at the university neither loses the rights nor escapes the responsibilities of citizenship. He or she is expected to obey federal, state, and local laws as well as the Regents' Rules, university regulations, and administrative rules. Students are subject to discipline for violating the standards of conduct whether such conduct takes place on or off campus, or whether civil or criminal penalties are also imposed for such conduct.

Academic Integrity

The faculty expects from its students a high level of responsibility and academic honesty. Because the value of an academic degree depends upon the absolute integrity of the work done by the student for that degree, it is imperative that a student demonstrate a high standard of individual honor in his or her scholastic work.

Scholastic dishonesty includes, but is not limited to, statements, acts or omissions related to applications for enrollment or the award of a degree, and/or the submission as one's own work or material that is not one's own. As a general rule, scholastic dishonesty involves one of the following acts: cheating, plagiarism, collusion and/or falsifying academic records. Students suspected of academic dishonesty are subject to disciplinary proceedings.

Plagiarism, especially from the web, from portions of papers for other classes, and from any other source is unacceptable and will be dealt with under the university's policy on plagiarism (see general catalog for details).

Email Use

The University of Texas at Dallas recognizes the value and efficiency of communication between faculty/staff and students through electronic mail. At the same time, email raises some issues concerning security and the identity of each individual in an email exchange. The university encourages all official student email correspondence be sent only to a student's U.T. Dallas email address and that faculty and staff consider email from students official only if it originates from a UTD student account. This allows the university to maintain a high degree of confidence in the identity of all individual corresponding and the security of the transmitted information. UTD furnishes each student with a free email account that is to be used in all communication with university personnel. The Department of Information Resources at U.T. Dallas provides a method for students to have their U.T. Dallas mail forwarded to other accounts.

Withdrawal from Class

The administration of this institution has set deadlines for withdrawal of any college-level courses. These dates and times are published in that semester's course catalog. Administration procedures must be followed. It is the student's responsibility to handle withdrawal requirements from any class. In other words, I cannot drop or withdraw any student. You must do the proper paperwork to ensure that you will not receive a final grade of "F" in a course if you choose not to attend the class once you are enrolled.

Student Grievance Procedures

Procedures for student grievances are found in Title V, Rules on Student Services and Activities, of the university's *Handbook of Operating Procedures*.

In attempting to resolve any student grievance regarding grades, evaluations, or other fulfillments of academic responsibility, it is the obligation of the student first to make a serious effort to resolve the matter with the instructor, supervisor, administrator, or committee with whom the grievance originates (hereafter called "the respondent"). Individual faculty members retain primary responsibility for assigning grades and evaluations. If the matter cannot be resolved at that level, the grievance must be submitted in writing to the respondent with a copy of the respondent's School Dean. If the matter is not resolved by the written response provided by the respondent, the student may submit a written appeal to the School Dean. If the grievance is not resolved by the School Dean's decision, the student may make a written appeal to the Dean of Graduate or Undergraduate Education, and the deal will appoint and convene an Academic Appeals Panel. The decision of the Academic Appeals Panel is final. The results of the academic appeals process will be distributed to all involved parties.

Copies of these rules and regulations are available to students in the Office of the Dean of Students, where staff members are available to assist students in interpreting the rules and regulations.

Incomplete Grade Policy

As per university policy, incomplete grades will be granted only for work unavoidably missed at the semester's end and only if 70% of the course work has been completed. An incomplete grade must be resolved within eight (8) weeks from the first day of the subsequent long semester. If the required work to complete the course and to remove the incomplete grade is not submitted by the specified deadline, the incomplete grade is changed automatically to a grade of **F**.

Disability Services

The goal of Disability Services is to provide students with disabilities educational opportunities equal to those of their non-disabled peers. Disability Services is located in room 1.610 in the Student Union. Office hours are Monday and Thursday, 8:30 a.m. to 6:30 p.m.; Tuesday and Wednesday, 8:30 a.m. to 7:30 p.m.; and Friday, 8:30 a.m. to 5:30 p.m.

The contact information for the Office of Disability Services is:
The University of Texas at Dallas, SU 22
PO Box 830688
Richardson, Texas 75083-0688
(972) 883-2098 (voice or TTY)

Essentially, the law requires that colleges and universities make those reasonable adjustments necessary to eliminate discrimination on the basis of disability. For example, it may be necessary to remove classroom prohibitions against tape recorders or animals (in the case of dog guides) for students who are blind. Occasionally an assignment requirement may be substituted (for example, a research paper versus an oral presentation for a student who is hearing impaired). Classes enrolled students with mobility impairments may have to be rescheduled in accessible facilities. The college or university may need to provide special services such as registration, note-taking, or mobility assistance.

It is the student's responsibility to notify his or her professors of the need for such an accommodation. Disability Services provides students with letters to present to faculty members to verify that the student has a disability and needs accommodations. Individuals requiring special accommodation should contact the professor after class or during office hours.

Religious Holy Days

The University of Texas at Dallas will excuse a student from class or other required activities for the travel to and observance of a religious holy day for a religion whose places of worship are exempt from property tax under Section 11.20, Tax Code, Texas Code Annotated.

The student is encouraged to notify the instructor or activity sponsor as soon as possible regarding the absence, preferably in advance of the assignment. The student, so excused, will be allowed to take the exam or complete the assignment within a reasonable time after the absence: a period equal to the length of the absence, up to a maximum of one week. A student who notifies the instructor and completes any missed exam or assignment may not be penalized for the absence. A student who fails to complete the exam or assignment within the prescribed period may receive a failing grade for that exam or assignment.

If a student or an instructor disagrees about the nature of the absence [i.e., for the purpose of observing a religious holy day] or if there is similar disagreement about whether the student has been given a reasonable time to complete any missed assignments or examinations, either the student or the instructor may request a ruling from the chief executive officer of the institution, or his or her designee. The chief executive officer or designee must take into account the legislative intent of TEC 51.911(b), and the student and instructor will abide by the decision of the chief executive officer or designee.

Off-Campus Instruction and Course Activities

Off-campus, out-of-state, and foreign instruction and activities are subject to state law and University policies and procedures regarding travel and risk-related activities. Information regarding these rules and regulations may be found at the website address given below. Additional information is available from the office of the school dean. (http://www.utdallas.edu/Business Affairs/Travel_Risk_Activities.htm)

These descriptions and timelines are subject to change at the discretion of Dr. Omar Hamdy.